CHAYAN GULGULIA

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Data Analyst | Business Analyst | SQL, Reporting & Reconciliation

Professional Experience

- **AdElement Media** — Pune, Maharashtra
  Data Analyst — *July 2025 – Oct 2026*
  - Grew a **$2.6M+ annual revenue portfolio** of 15 partner accounts by **32% YoY** through request filtering and floor-price tests, and scaled two new accounts about 10x
  - Cut the **month-end close from 14 days to 3–4** for company-wide billing (55+ revenue lines, 10+ entities) by building SQL views that put our revenue and partner revenue from several databases side by side
  - Found a **bug in the revenue-change formula** that misreported daily changes for all 70+ demand partners, capping real jumps at 100%; fixed it, proved the fix with UAT against source data, and had it confirmed by the data team head
  - Reconciled **100K+ records a month** against partner reports in SQL (Redshift, AWS Athena, MySQL) before each invoice went out, fixing time-zone mismatches and flagging every gap above 5%
  - Opened **new revenue from low-volume publishers**: analyzed their bundles and found new opportunities; leadership started a few, and they later became strong earners
  - Replaced manual daily reports for 10+ accounts with a SQL, Python and Excel (Power Query, VBA) pipeline, saving **7–8 hours a week**, and built 3 Metabase dashboards that flagged IVT spikes and revenue drops without a query
  - Saved **about an hour a day** by rewriting a SQL query on TB-scale logs (**20 minutes to 10** per account): joined the data inside the query, filtered early and dropped repeated fields
  - Saved **2–3 hours a day** on the slowest part of analysis, pulling and checking data, by connecting **Claude to internal databases through MCP** (read-only, company-approved), with every answer verified against source
  - Turned around **3+ ad-hoc leadership requests a week within 2–3 hours**, and presented the monthly revenue report to the CEO, COO, founder and parent-company executives
- **Datawrkz** — Bengaluru, Karnataka
  Ad Operations Analyst Intern — *Mar 2025 – Jun 2025*
  - Kept 8 campaigns (**£160K budget**) within 95–105% of planned spend by adjusting CPM bids from daily Excel pacing reports; automated the reporting through APIs and Looker, cutting **turnaround 35%**
  - Lifted **conversions 15–23%** (sign-ups, booked calls) for 8+ real-estate clients by splitting audiences into buyer groups, such as high-net-worth and retirement investors, and A/B testing them

Projects

- **Tallyline: Ad Exchange Analytics Workspace** | *Python, SQL, DuckDB, FastAPI* — Live demo | GitHub | *2026*
  - Built an analytics workspace for a simulated ad marketplace: KPIs, a full bid funnel, six drill-down levels and printable reports, on a DuckDB star schema behind a read-only FastAPI
  - Wrote an **anomaly detector in SQL** that caught **100% of planted incidents** at 94–100% precision, with 31 tests checking every breakdown adds back up to the total
- **Credit Risk & Loan Default Analytics** | *Snowflake SQL, XGBoost, Tableau* — Tableau | GitHub | *2026*
  - Analyzed **150K+ loan records** in Snowflake SQL (CTEs, window functions, cohorts): borrowers with high card utilization and past late payments defaulted at 48%, against 3% for low-risk borrowers
  - Compared logistic regression and XGBoost (test AUC 0.86 and 0.87) and turned the scores into a lending rule: declining the riskiest 20% catches **73% of defaulters**; shared in two Tableau dashboards
- **Quick-Commerce Funnel Analytics** | *dbt, PostgreSQL, Python, Metabase* — Case study | GitHub | *2026*
  - Modeled **120K Amazon India orders** and Blinkit orders, customers and campaigns in **dbt on PostgreSQL** (25 models, 58 data tests), with funnel, cohort and RFM views in Metabase
  - Ran a **simulated A/B test** of a Rs 100 win-back offer: reorders rose 10.1% (95% CI 3.1–17.2%), but each extra reorder cost Rs 1,087 against Rs 609 margin, so recommended not rolling it out
  - Found that four eastern states **return orders at twice the rate** of the rest of India (12.5% vs. 6.4%), and kept a promotion field out because it is only filled in after checkout
- **Nifty 50 Risk, Regimes & Backtests** | *DuckDB SQL, Python, Tableau* — Tableau | GitHub | *2026*
  - Backtested trading rules and 5 portfolio weightings on **Nifty 50 stocks** out of sample (2017–2026, after costs): a moving-average timing rule earned **6.6% a year against 12.4%** for buy-and-hold
  - Tested one-day **Value at Risk (VaR)** three ways with Kupiec and Christoffersen backtests; only **GARCH-t passed both at 99%** for the index (36 breaches against 26.6 expected)
  - Showed that momentum's **25.6% a year** was mostly **survivorship bias**: equal weight on the same 50 stocks made 21.9%
- **Movie Recommender on Databricks** | *PySpark, Spark MLlib, Databricks* — Notebooks | GitHub | *2026*
  - Built a **Spark ALS recommender** and scaled it from 100K to **25M ratings**; it beat a popularity baseline by **94% on NDCG@10**, tested on what each user watched next
  - Measured how it scales on serverless Spark: **100x more data took 8.3x longer** to train (41 s to 339 s)

Technical Skills

- **Data**: SQL (PostgreSQL, Redshift, AWS Athena, Snowflake, DuckDB), Python (Pandas, NumPy, Scikit-learn), dbt, PySpark
- **BI & Tools**: Excel (Power Query, Pivot Tables, XLOOKUP, VBA), Power BI (DAX), Tableau, Metabase, Looker, Jira, Git, AWS

Education

- **National Institute of Technology, Silchar** — Assam, India
  B.Tech in Civil Engineering, CGPA 8.24 — *2021 – 2025*

Certifications & Leadership

- **Certifications**: Data Visualization with Power BI (Great Learning, 2024); Data Analysis using Microsoft Excel (Coursera, 2024)
- **Leadership**: General Secretary, NITSMUN (2024–25), participation up 70%; Executive Head, E-Cell (2023–24), 10 workshops